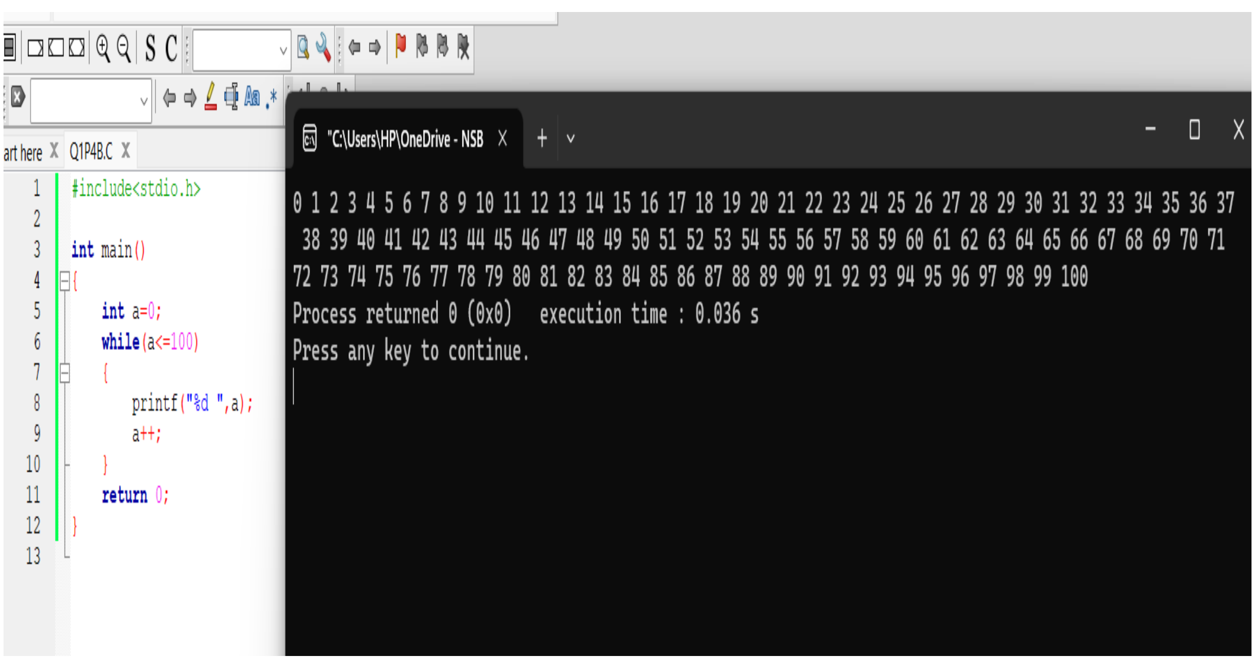
PRACTICAL 5

Name : Ranudi Gayathmie Kariyapperuma
Index : 29897
Batch No : 23.1
Degree Program Name : Technology Management

Section A

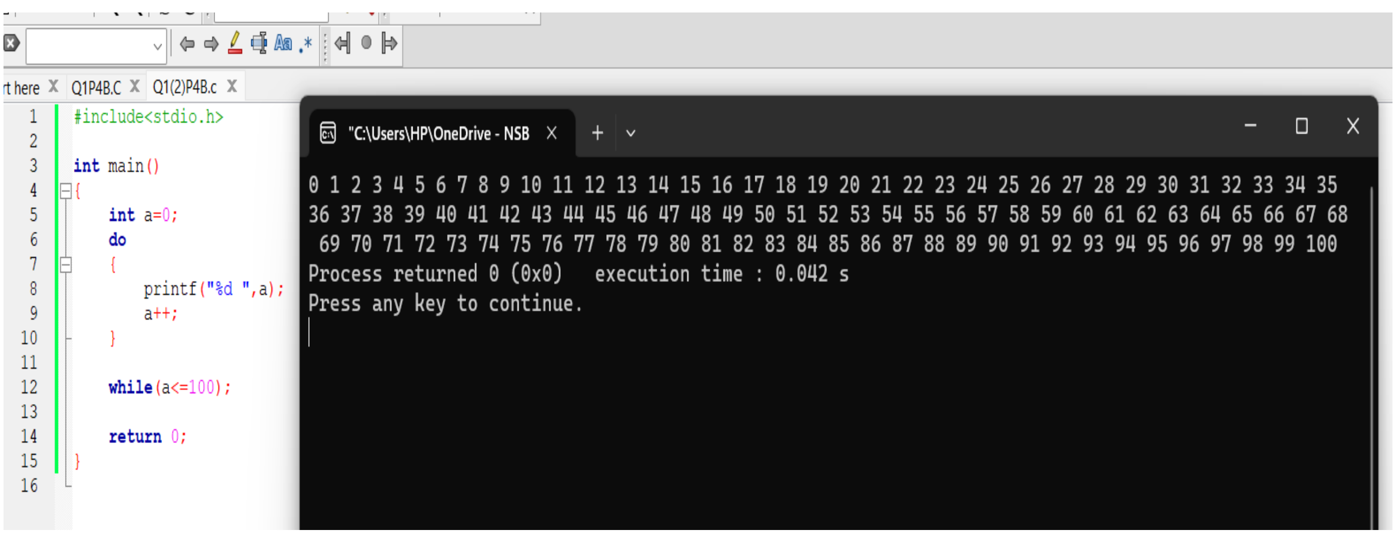
(Q1) - **While Loop**

```
int main()
{
  int a=0;
  while(a<=100)
    {
       printf("%d ",a);
       a++;
     }
  return 0;
 }
```



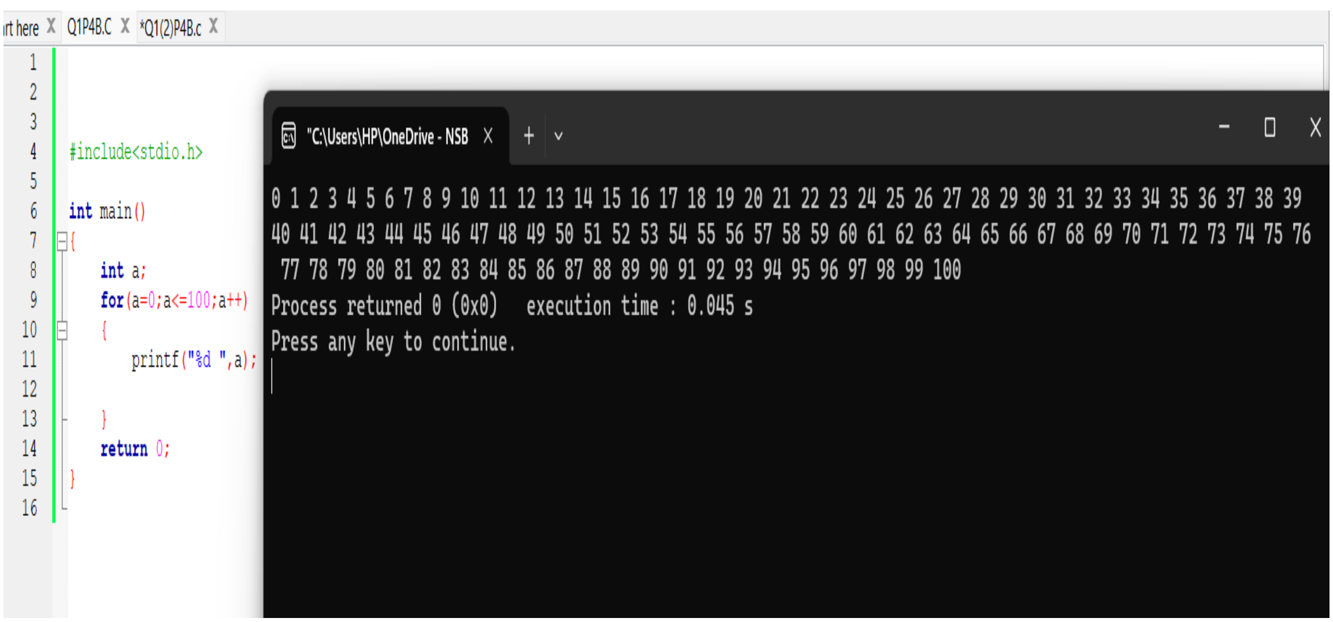
- **Do while**

```
#include<stdio.h>

int main()
{
     int a=0;
     do
     {
          printf("%d ",a);
          a++;
     }

     while(a<=100);

     return 0;
}
```

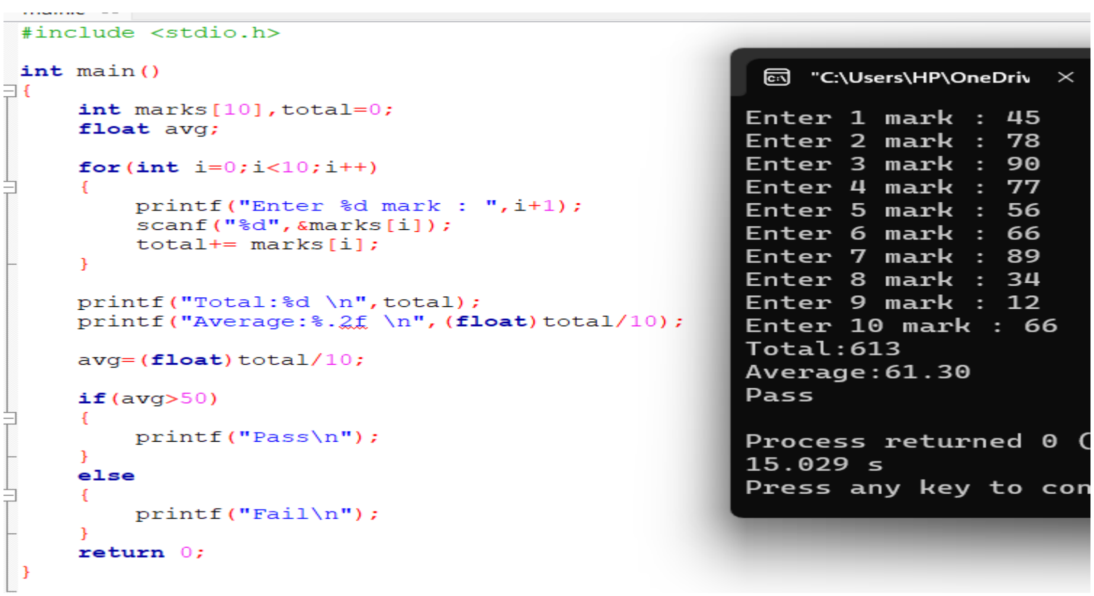


- **For Loop**

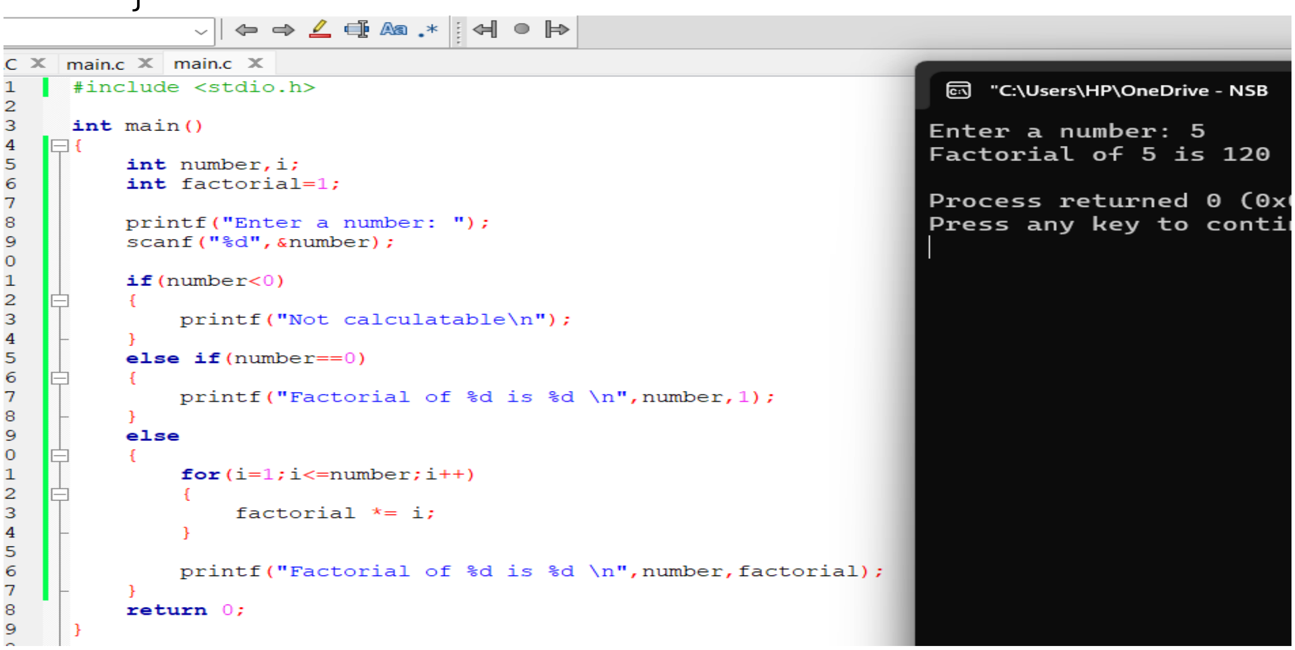
```
#include<stdio.h>

int main()
{
    int a;
    for(a=0;a<=100;a++)
    {
        printf("%d ",a);

    }
    return 0;
}
```



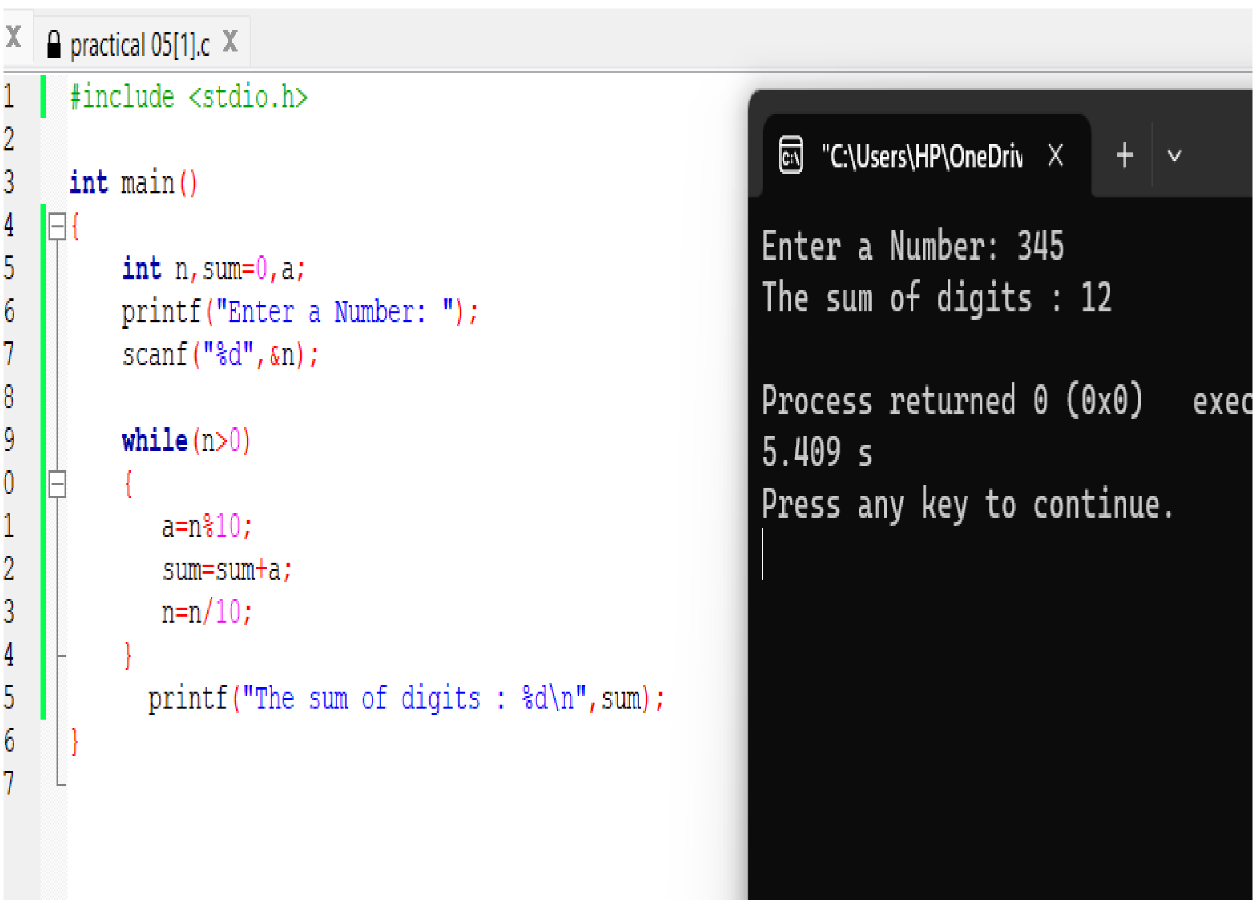
Q2)
```c
#include <stdio.h>
int main()
{
    int marks[10],total=0;
    float avg;

    for(int i=0;i<10;i++)
    {
        printf("Enter %d mark : ",i+1);
        scanf("%d",&marks[i]);
        total+= marks[i];
    }
    printf("Total:%d \n",total);
    printf("Average:%.2f \n",(float)total/10);

    avg=(float)total/10;

    if(avg>50)
    {
        printf("Pass\n");
    }
    else
    {
        printf("Fail\n");
    }
    return 0;
}
```



Q3)
```
#include <stdio.h>
int main()
{
    int number,i;
    int factorial=1;

    printf("Enter a number: ");
    scanf("%d",&number);
     if(number<0)
    {
        printf("Not calculatable\n");
    }
    else if(number==0)
    {
        printf("Factorial of %d is %d \n",number,1);
    }
    else
    {
        for(i=1;i<=number;i++)
        {
            factorial *= i;
        }
      printf("Factorial of %d is %d \n",number,factorial);
    }
    return 0;
}
```

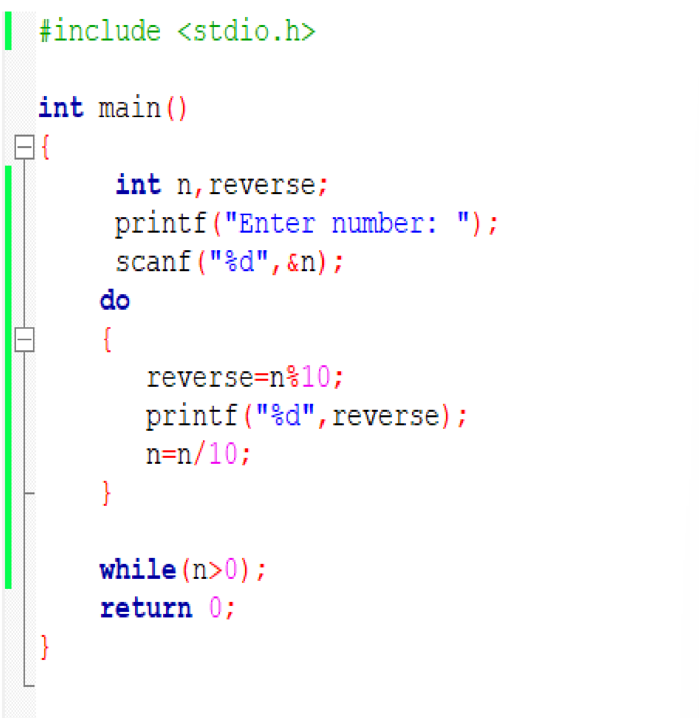


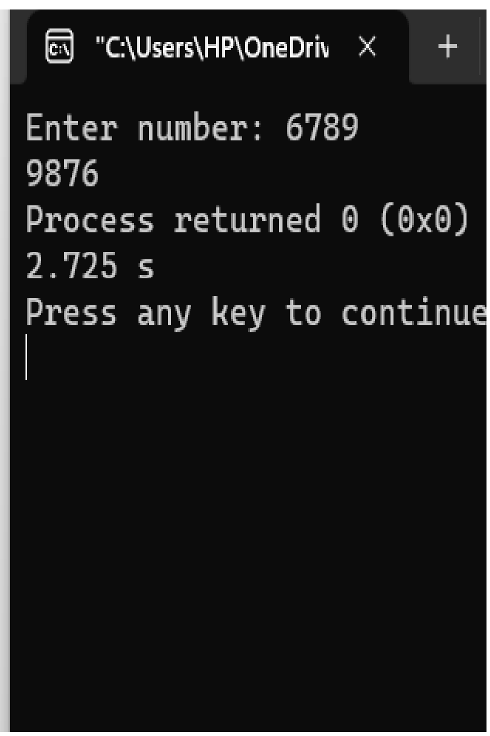
Q4)
```
#include <stdio.h>
int main()
{
     int n,sum=0,a;
     printf("Enter a Number: ");
     scanf("%d",&n);

     while(n>0)
     {
        a=n%10;
        sum=sum+a;
        n=n/10;
     }
       printf("The sum of digits : %d\n",sum);
}
```



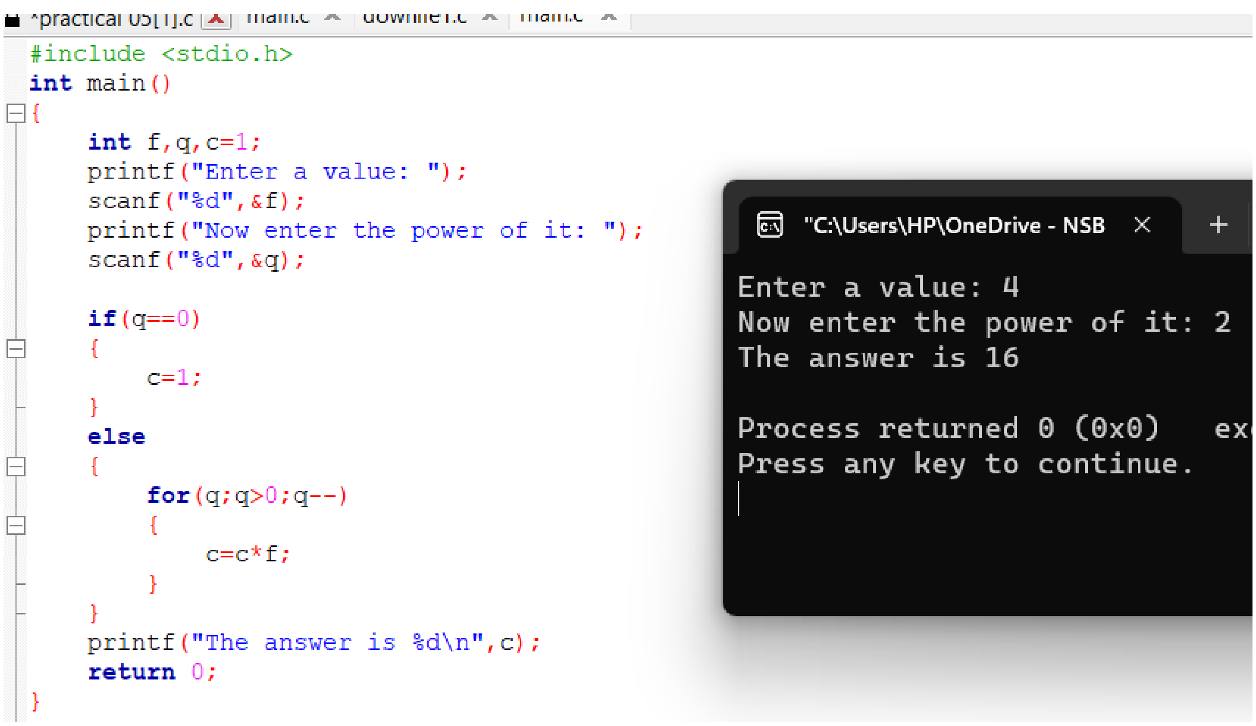
Q5)

```
#include <stdio.h>
int main()
{
     int n,reverse;
     printf("Enter number: ");
     scanf("%d",&n);
    do
    {
       reverse=n%10;
       printf("%d",reverse);
       n=n/10;
    }

    while(n>0);
    return 0;
}
```

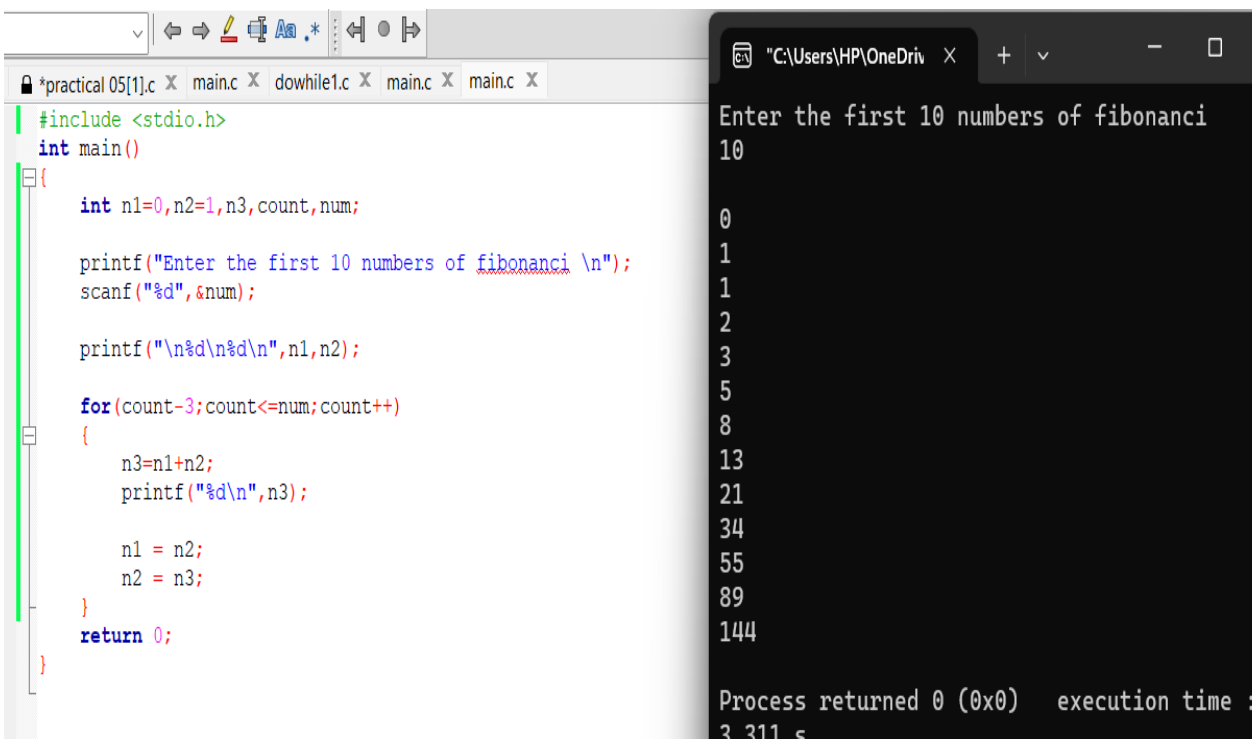




Q6)
```
#include <stdio.h>
int main()
{
    int f,q,c=1;
    printf("Enter a value: ");
    scanf("%d",&f);
    printf("Now enter the power of it: ");
    scanf("%d",&q);

    if(q==0)
    {
        c=1;
    }
    else
    {
        for(q;q>0;q--)
        {
            c=c*f;
        }
    }
    printf("The answer is %d\n",c);
    return 0;
}
```

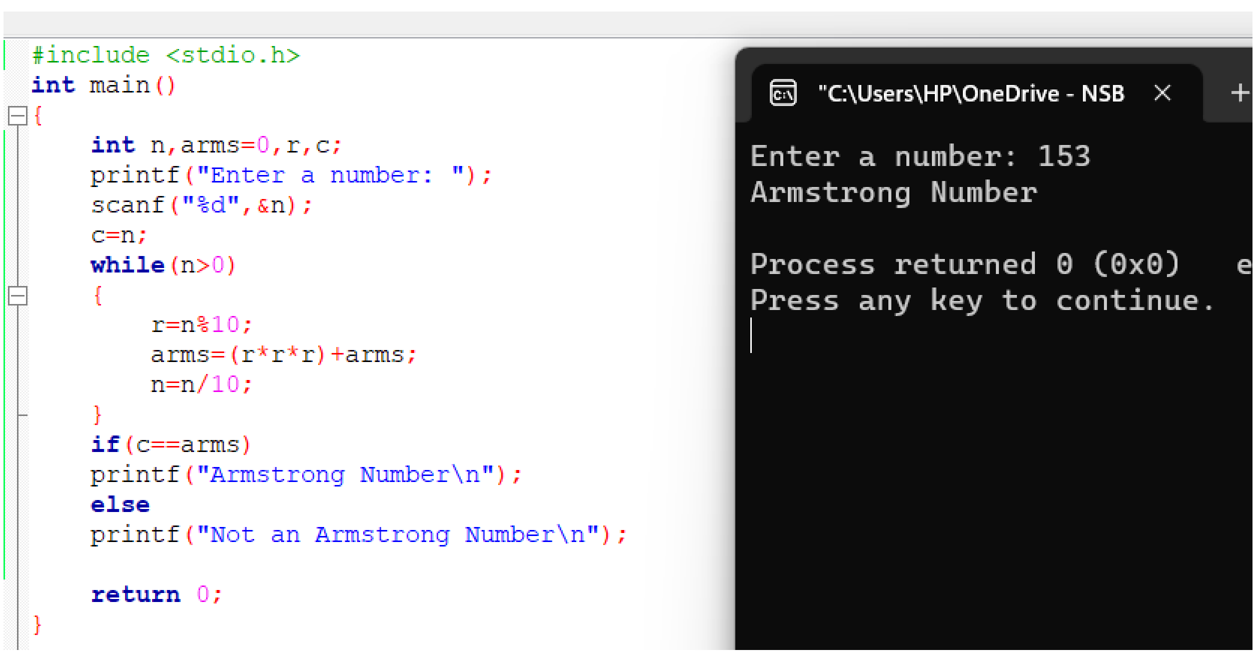


Q7)
```
#include <stdio.h>
int main()
{
    int n1=0,n2=1,n3,count,num;

    printf("Enter the first 10 numbers of fibonanci \n");
    scanf("%d",&num);

    printf("\n%d\n%d\n",n1,n2);

    for(count-3;count<=num;count++)
    {
        n3=n1+n2;
        printf("%d\n",n3);

        n1 = n2;
        n2 = n3;
    }
    return 0;
}
```



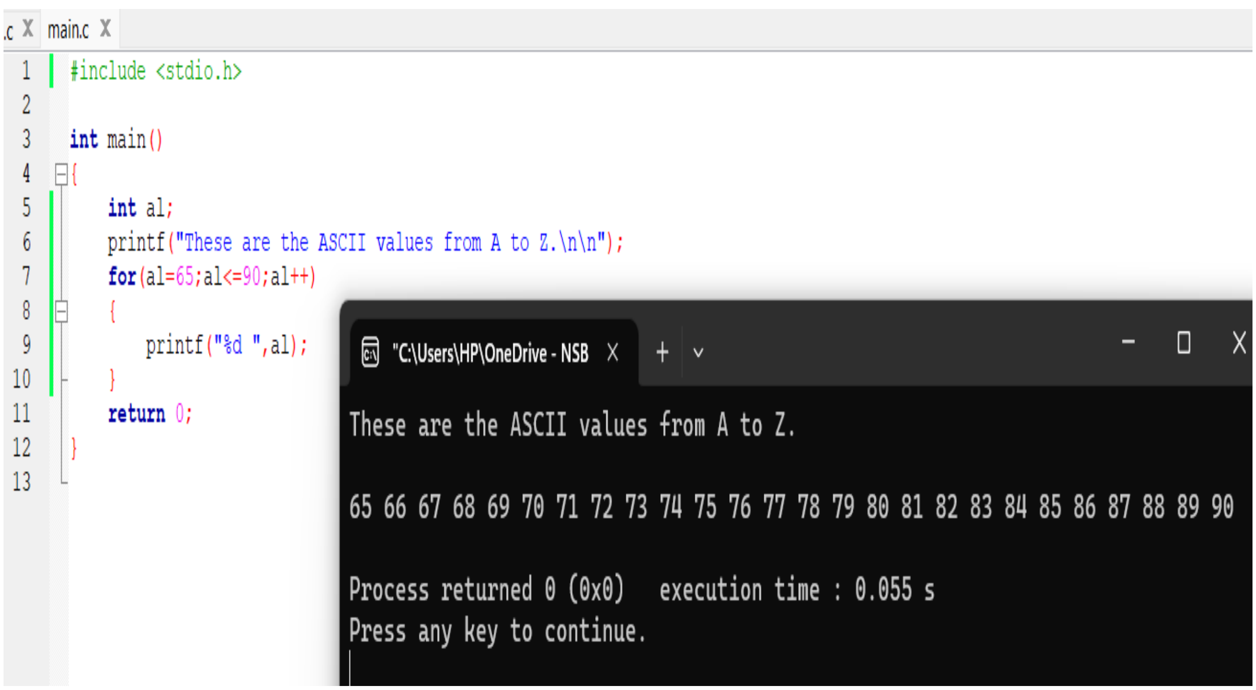
Q8)

```
#include <stdio.h>
int main()
{
    int n,arms=0,r,c;
    printf("Enter a number: ");
    scanf("%d",&n);
    c=n;
    while(n>0)
    {
        r=n%10;
        arms=(r*r*r)+arms;
        n=n/10;
    }
    if(c==arms)
    printf("Armstrong Number\n");
    else
    printf("Not an Armstrong Number\n");

    return 0;
}
```



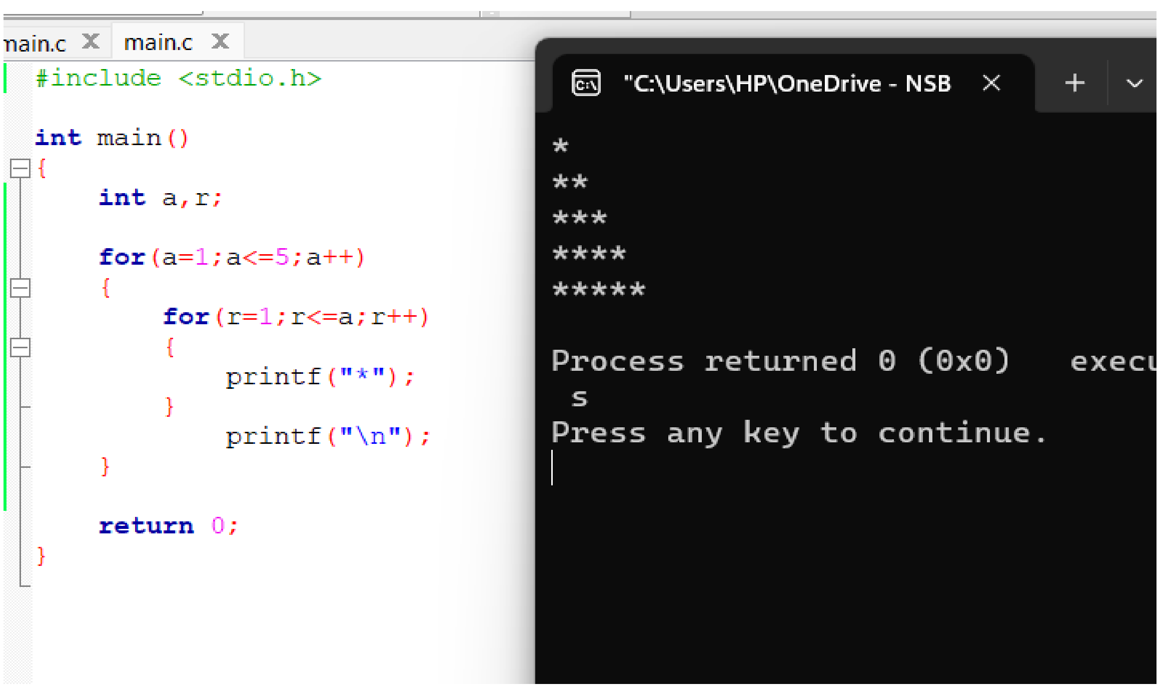
Q9)

```c
#include <stdio.h>

int main()
{
    int al;
    printf("These are the ASCII values from A to Z.\n\n");
    for(al=65;al<=90;al++)
    {
        printf("%d ",al);
    }
    return 0;
}
```



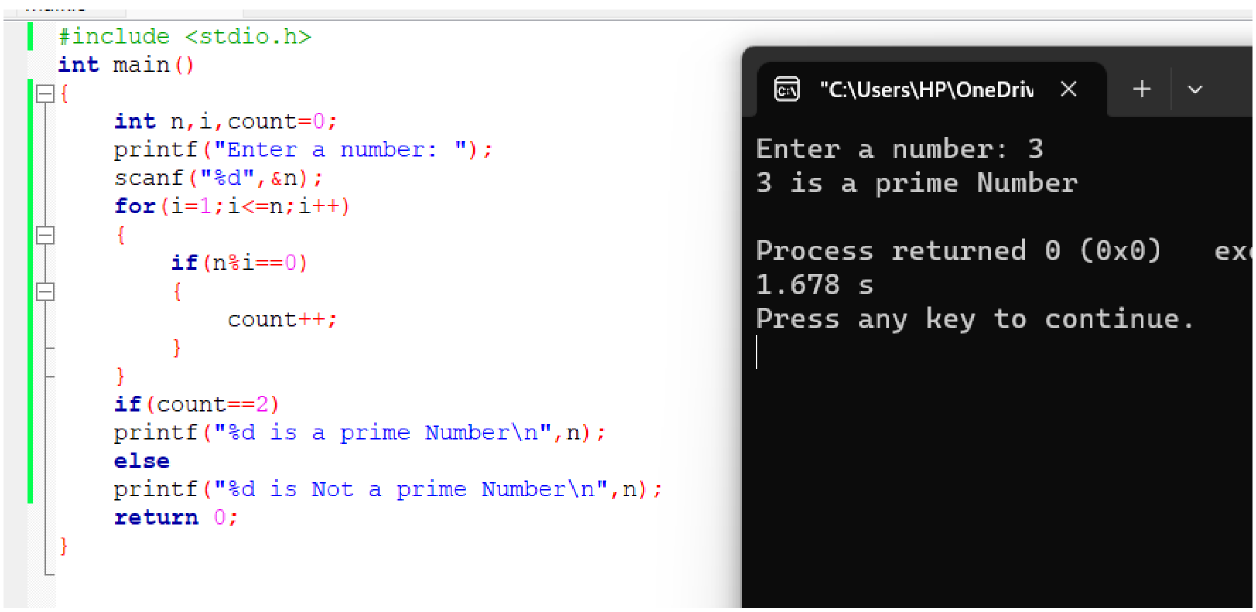
Q10)

```c
#include <stdio.h>
int main()
{
    int a,r;
    for(a=1;a<=5;a++)
    {
        for(r=1;r<=a;r++)
        {
            printf("*");
        }
            printf("\n");
    }
    return 0;
}
```



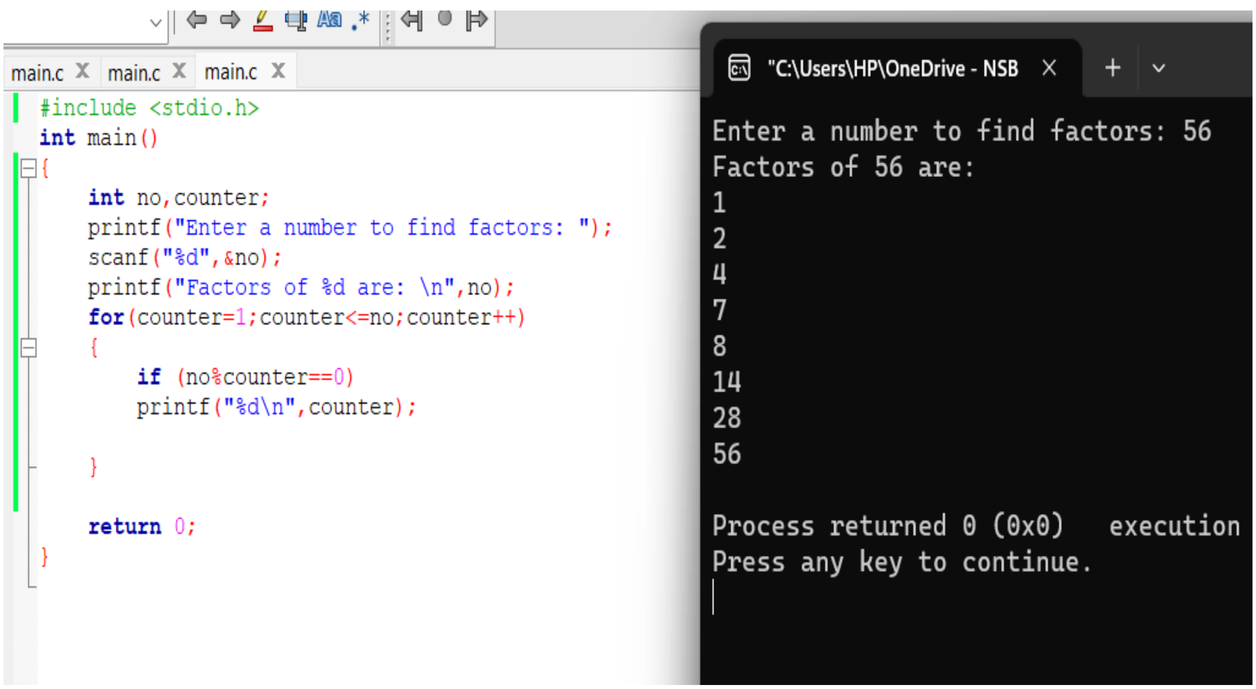
Q11)

```c
#include <stdio.h>
int main()
{
    int n,i,count=0;
    printf("Enter a number: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
        }
    }
    if(count==2)
    printf("%d is a prime Number\n",n);
    else
    printf("%d is Not a prime Number\n",n);
    return 0;
}
```



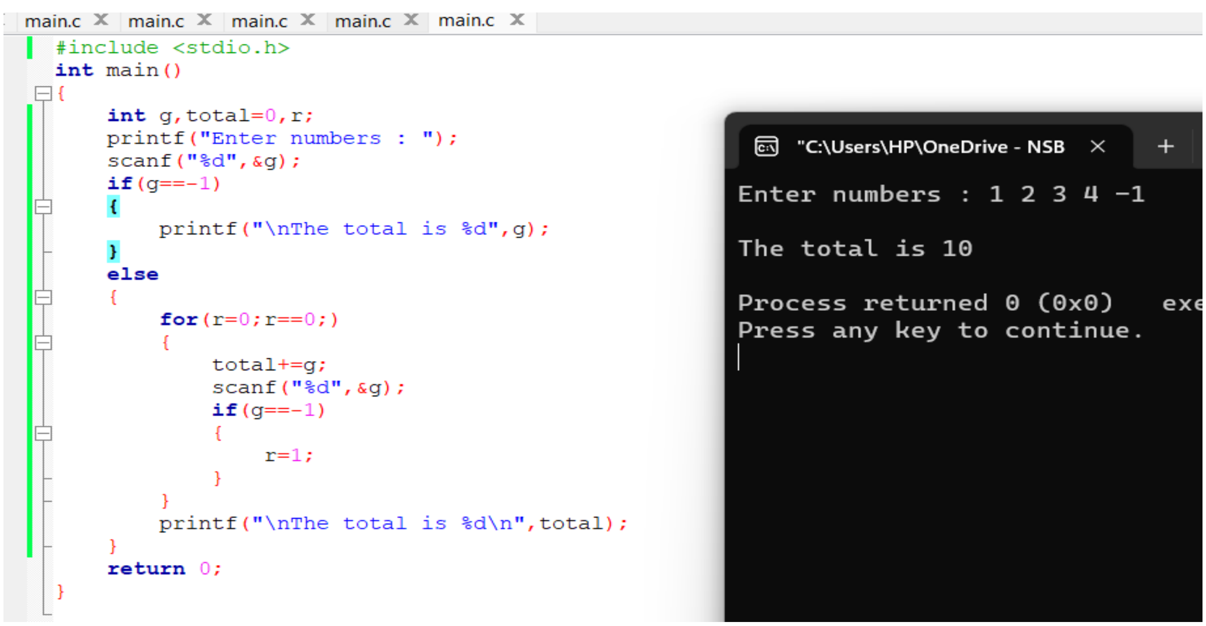
Q12)

```
#include <stdio.h>
int main()
{
    int no,counter;
    printf("Enter a number to find factors: ");
    scanf("%d",&no);
    printf("Factors of %d are: \n",no);
    for(counter=1;counter<=no;counter++)
    {
        if (no%counter==0)
        printf("%d\n",counter);

    }

    return 0;
}
```



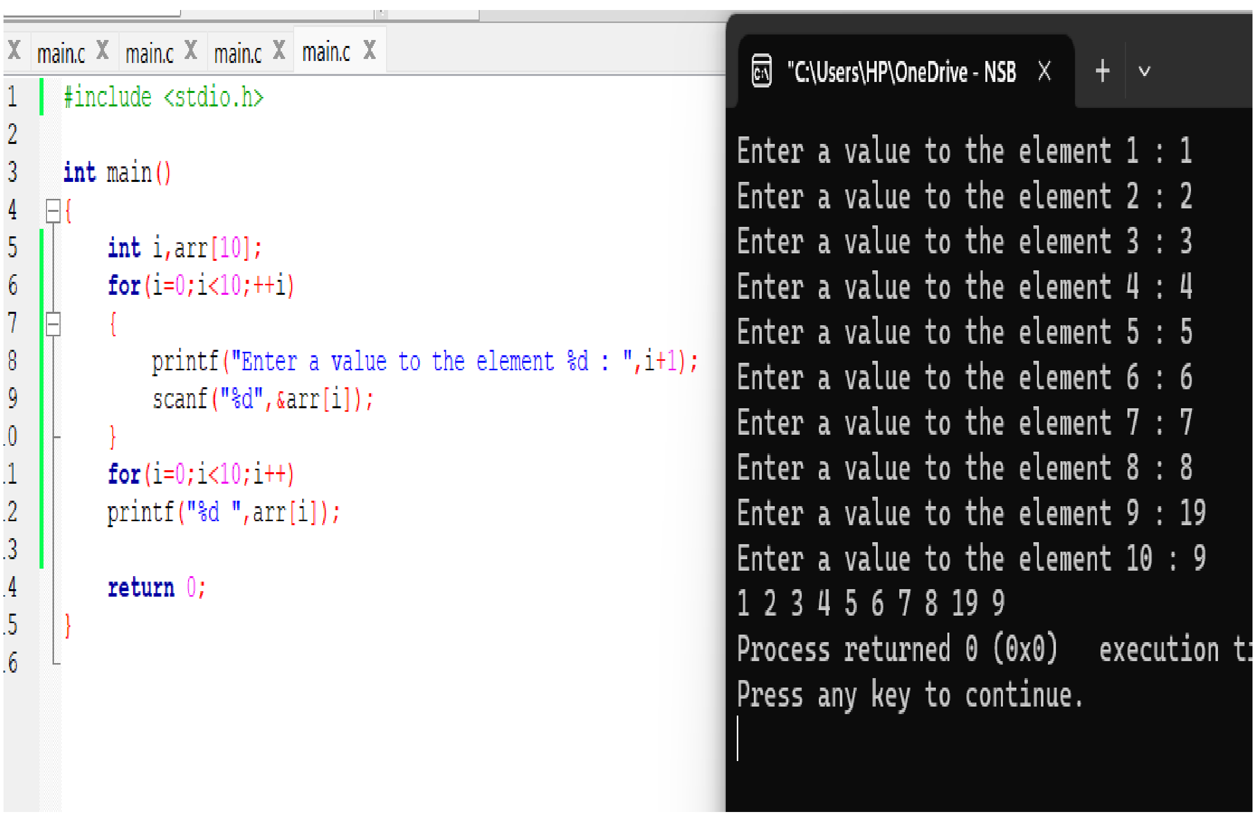
Q12)

```c
#include <stdio.h>
int main()
{
    int g,total=0,r;
    printf("Enter numbers : ");
    scanf("%d",&g);
    if(g==-1)
    {
        printf("\nThe total is %d",g);
    }
    else
    {
        for(r=0;r==0;)
        {
            total+=g;
            scanf("%d",&g);
            if(g==-1)
            {
                r=1;
            }
        }
        printf("\nThe total is %d\n",total);
    }
    return 0;
}
```



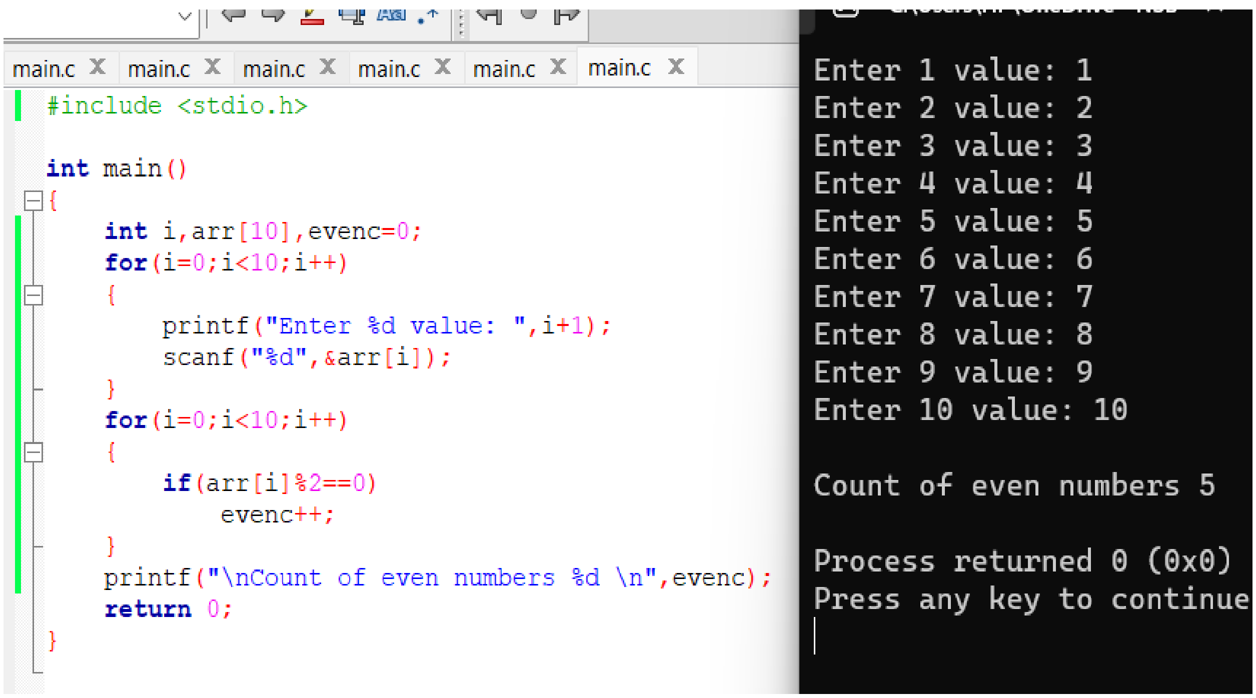
Q13)

```c
#include <stdio.h>
int main()
{
    int i,arr[10];
    for(i=0;i<10;++i)
    {
        printf("Enter a value to the element %d : ",i+1);
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    printf("%d ",arr[i]);

    return 0;
}
```



Q14)

```c
#include <stdio.h>
int main()
{
    int i,arr[10],evenc=0;
    for(i=0;i<10;i++)
    {
        printf("Enter %d value: ",i+1);
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    {
        if(arr[i]%2==0)
            evenc++;
    }
    printf("\nCount of even numbers %d \n",evenc);
    return 0;
}
```



Section B

Q1)

```
#include <stdio.h>
int main()
{
    {int no,counter=0,negative=0,positive=0,zero=0;

    for(counter;counter<10;counter++)
    {
        printf("Enter number : ");
        scanf("%d",&no);
        if(no==0)
        {
            zero++;
        }
        else if(no<0)
        {
            negative++;
        }
        else
        {
            positive++;
        }
    }
   printf("There are \n %d positive numbers \n %d negative number
          \n %d Zeros",positive,negative,zero);
    return 0;
    }
  }
```

```c
#include <stdio.h>
int main()
{
    {int no,counter=0,negative=0,positive=0,zero=0;

    for(counter;counter<10;counter++)
    {
        printf("Enter number : ");
        scanf("%d",&no);
        if(no==0)
        {
            zero++;
        }
        else if(no<0)
        {
            negative++;
        }
        else
        {
            positive++;
        }
    }
   printf("There are \n %d positive numbers \n %d negative number \n %d Zeros",positive,negative,zero);
    return 0;
    }
  }
```

"C:\Users\HP\OneDrive - NSB

```
Enter number : 4
Enter number : 5
Enter number : 0
Enter number : -34
Enter number : 6
Enter number : 7
Enter number : 79
Enter number : -9
There are
 6 positive numbers
 3 negative number
 1 Zeros
Process returned 0 (0x0)   execution t
 s
Press any key to continue.
```

Q2)

```c
#include <stdio.h>
int main()
{
    int i,marks[10],max,min,total=0;
    float avg;
    printf("Enter the marks of 10 students\n\n");

    max=marks;
    min=marks;
    for(i=0;i<10;i++)
    {
        printf("Enter marks for student %d: ",i+1);
        scanf("%d",&marks[i]);
        total+=marks[i];
        if(i==0||marks[i]>max)
        {
            max=marks[i];
        }

        if(min>marks[i])
        {
            min=marks[i];
        }
    }
    avg=(float)total/10;

    printf("\nThe maximum number is %d\n",max);
    printf("The minimum number is %d\n",min);
    printf("The average is %2f\n",avg);
    return 0;
}
```

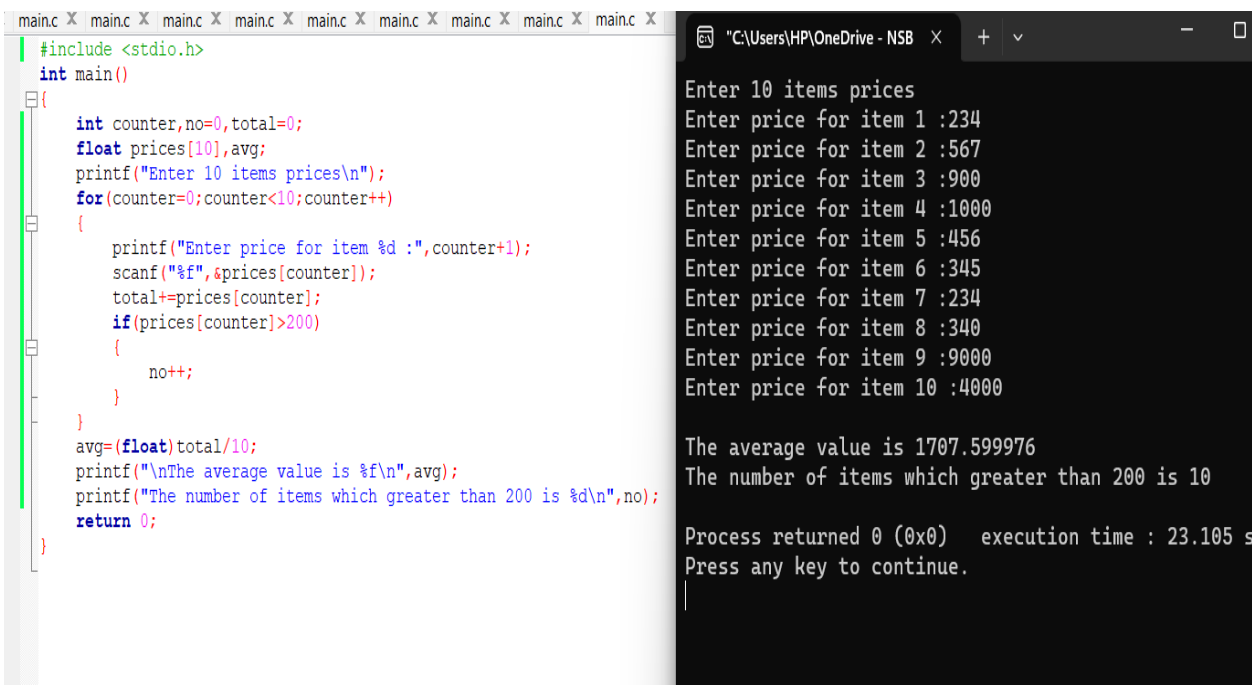
```c
#include <stdio.h>
int main()
{
    int i,marks[10],max,min,total=0;
    float avg;
    printf("Enter the marks of 10 students\n\n");

    max=marks;
    min=marks;
    for(i=0;i<10;i++)
    {
        printf("Enter marks for student %d: ",i+1);
        scanf("%d",&marks[i]);
        total+=marks[i];
        if(i==0||marks[i]>max)
        {
            max=marks[i];
        }

        if(min>marks[i])
        {
            min=marks[i];
        }
    }
    avg=(float)total/10;

    printf("\nThe maximum number is %d\n",max);
    printf("The minimum number is %d\n",min);
    printf("The average is %2f\n",avg);
    return 0;
}
```

"C:\Users\HP\OneDrive - NSB

```
Enter the marks of 10 students

Enter marks for student 1: 34
Enter marks for student 2: 56
Enter marks for student 3: 78
Enter marks for student 4: 90
Enter marks for student 5: 23
Enter marks for student 6: 22
Enter marks for student 7: 66
Enter marks for student 8: 56
Enter marks for student 9: 45
Enter marks for student 10: 12

The maximum number is 90
The minimum number is 12
The average is 48.200001

Process returned 0 (0x0)   execution
Press any key to continue.
```

Q3)

```c
#include <stdio.h>
int main()
{
    int counter,no=0,total=0;
    float prices[10],avg;
    printf("Enter 10 items prices\n");
    for(counter=0;counter<10;counter++)
    {
        printf("Enter price for item %d :",counter+1);
        scanf("%f",&prices[counter]);
        total+=prices[counter];
        if(prices[counter]>200)
        {
            no++;
        }
    }
    avg=(float)total/10;
    printf("\nThe average value is %f\n",avg);
    printf("The number of items which greater than 200 is %d\n",no);
    return 0;
}
```



Q4)

```
#include <stdio.h>
int main()
{
    int empno,counter=0;
    float basicsalary;
    printf("Enter the employee number and basicsalary(enter -999 to
stop employee number) \n");

    while(1)
    {   printf("\nEmployee no: ");
        scanf("%d",&empno);

        if(empno==-999)
        {
            break;
        }
        printf("Basic Salary: ");
        scanf("%f",&basicsalary);

        if(basicsalary>=5000)
        {
            counter++;
        }
    }
    printf("\nThe number of employees whose Basic Salary >=5000 is
           %d\n",counter);
    return 0;
}
```

```c
#include <stdio.h>
int main()
{
    int empno,counter=0;
    float basicsalary;
    printf("Enter the employee number and basicsalary(enter -999 to stop employee number) \n");

    while(1)
    {   printf("\nEmployee no: ");
        scanf("%d",&empno);

        if(empno==-999)
        {
            break;
        }
        printf("Basic Salary: ");
        scanf("%f",&basicsalary);

        if(basicsalary>=5000)
        {
            counter++;
        }
    }
    printf("\nThe number of employees whose Basic Salary >=5000 is %d\n",counter);
    return 0;
}
```

```
Enter the employee number and basicsalary(enter
 -999 to stop employee number)

Employee no: 1
Basic Salary: 20000

Employee no: 2
Basic Salary: 90000

Employee no: 3
Basic Salary: 50000

Employee no: -999

The number of employees whose Basic Salary >=50
00 is 3

Process returned 0 (0x0)   execution time : 16.
704 s
```

Q5)
```
#include <stdio.h>
int main()
{
    int empno,workhours,overtpayment,counter=0,totalemp=0;
    printf("Enter your employee number hours worked(enter-999 to
stop employee number)\n");

    while(1)
    {
        printf("\nEnter Employee No: ");
        scanf("%d",&empno);

        if(empno==-999)
        {
            break;
        }
        printf("Enter Hours Work: ");
        scanf("%d",&workhours);

        totalemp++;

        if(workhours<=40)
        {
            overtpayment=0;
        }
        else
        {
            overtpayment=workhours-40;
            overtpayment=(overtpayment*150)+((40-workhours)*150);
        }

        if(overtpayment)
        printf("Employee No: ",empno);
        printf("overtimepayment: ",overtpayment);

        if(overtpayment>=4000)
        {
            counter++;
        }

        float percentage=((float)counter/totalemp)*100;
```

```
        printf("Percenntage of employees with overtime payment
 exceeding Rs4000 : %2f ",percentage);
    }
    return 0;
}
```